

Scientific Reviewer

Strengths · weaknesses · verdict · 13+3

VERDICT PREVIEW

Accept-as-systems-WFM; place as generative world simulator, not latent planner SOTA. Demand precise claims: synth-data vs online planning.

SPINE: Generative video as physics simulator — Predict2.5 (flow T2W/I2W/V2W) + Reason1 + Transfer2.5 | PARALLEL to latent WMs (Dreamer/JEPA/LeWM)
| synth data · policy eval · closed-loop robotics/AV

Claims under review

What the paper asks us to believe

C1 Unified flow-matching Predict2.5 is a strong Physical AI video WFM (PAI $\approx$ Wan-class).

C2 Transfer2.5 is 3.5 $\times$ smaller than Transfer1 yet higher fidelity + long-horizon RNDS.

C3 Real2Real Transfer aug lifts Diffusion Policy 5/30 $\rightarrow$ 24/30 under visual OOD.

C4 Suite enables synth data, policy eval, closed-loop hooks for robotics/AV.

C5 Pixel generative WFM is a distinct paradigm from latent WMs (Ha/Dreamer/V-JEPA 2).

Strengths

What holds under scrutiny

Open WFM narrative

Coherent Physical AI story + open code/ckpts (NVIDIA Open Model License).

Broad eval suite

PAI-Bench auto + human prefs + Transfer + robot hardware + AV detectors + DreamGen + Bridge AC.

Action inject factorial

Tab.20 TimeEmb $\gg$ Cross $\gg$ Concat — clean within-family ablate.

Concrete method upgrades

FM, unified conditioning, Reason1, domain soup, RL, distillation — not vague “scale.”

Honest paradigm split

§7 cites V-JEPA 2 correctly as other (latent) pole — not genealogy theft.

Distill near-parity

4-step student $\approx$ teacher on PAI Overall (0.768 $\rightarrow$ 0.764 T2W).

Weaknesses / critique

Demand precision

Systems density

Ablations often qualitative (FM vs EDM without matched retrain table).

Robot n small

3 trials × 10 scenarios (30); single kitchen bimanual apple-bowl task.

DreamGen ≠ VLA SR

Tab.18 = video instruction-following; risk of over-reading as robot success.

Closed-loop framing

Long-horizon Transfer + AC rollouts ≠ full agent-in-loop SR curves.

Proprietary data

3.1M driving clips + filtering VLMs — data reproducibility limited.

Wan fairness

Size mismatches; human votes help but ≠ matched compute bake-off.

PAI-Bench evidence (Tabs. 10-11)

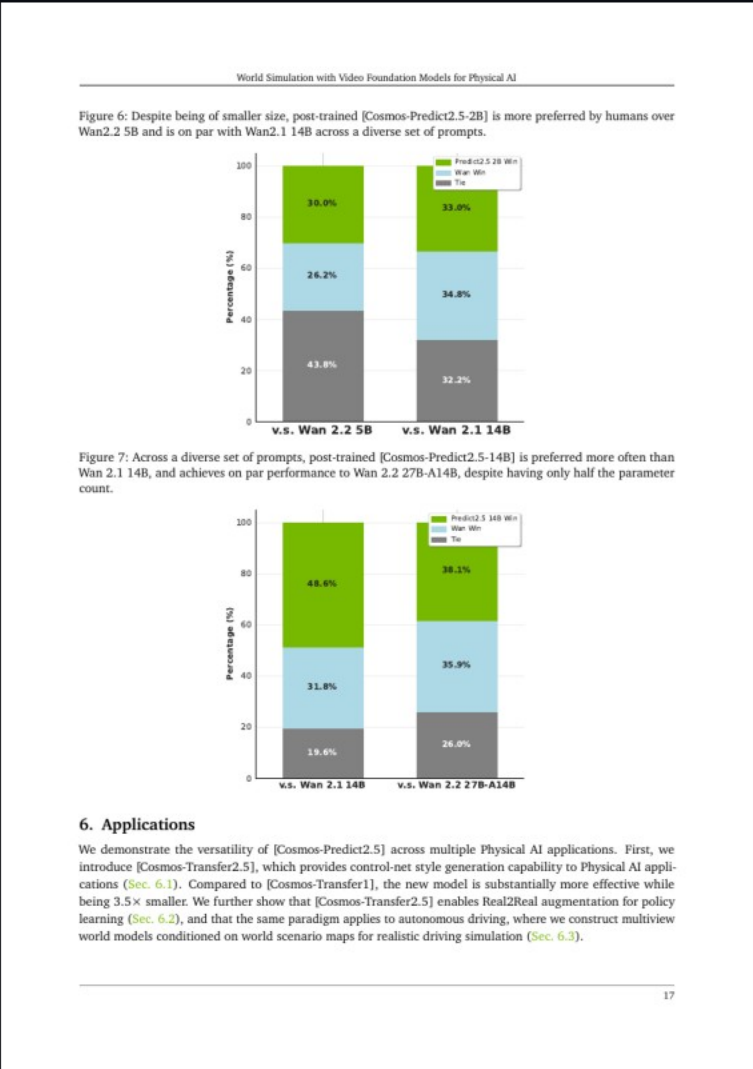
Post-train 2B/14B tie Overall; I2W best-in-table 0.810

Model	T2W Overall	I2W Overall	
Predict2.5-2B post	0.768	0.810	★
Predict2.5-14B post	0.768	0.810	★
Wan2.1-14B	0.761	0.797	
Wan2.2-5B	0.764	0.804	
Wan2.2-27B-A14B	0.769	0.806	T2W Domain edge

Caveat: Wan2.2-27B still edges T2W Domain (0.810 vs 0.804). Human prefs (Fig.7) show clearer 14B benefit than auto Overall ties.

Fig. 6-7 human prefs (reviewer lens)

Human taste ≠ automatic Overall — 14B jump vs same Wan2.1-14B (33%→48.6%)



Paper Fig. 7 — prefer human+auto jointly; size still matters for taste.

Robot claim precision (Tab. 13)

CLEAR number · narrow scope

1/30

Base

5/30

Image augs

24/30

Transfer2.5

Reviewer demands

- Single task (apple→bowl), 3 seeds/cell — suggestive, not multi-task SOTA.
- Mechanism = offline visual aug for Diffusion Policy — NOT online world-model MPC.
- Do not upgrade to “closed-loop autonomy solved.”
- Confound: 5× offline synth variants vs on-the-fly photometric baseline.

DreamGen & Bridge — claim hygiene

IF $\neq$ SR · video metrics $\neq$ control success

DreamGen Tab. 18

Object GPT 91.8 — video instruction-following (VLM judges).

Pipeline to IDM/latent actions described — hardware VLA SR NOT the table metric.

PUNISH reading Tab.18 as BEHAVIOR/VLA success.

Bridge AC Tab. 19-20

PSNR 24.95 / FVD 146 vs Predict1 21.14 / 190.

Suited for policy evaluation (paper claim); closed-loop control SR with AC as planner not separately tabulated.

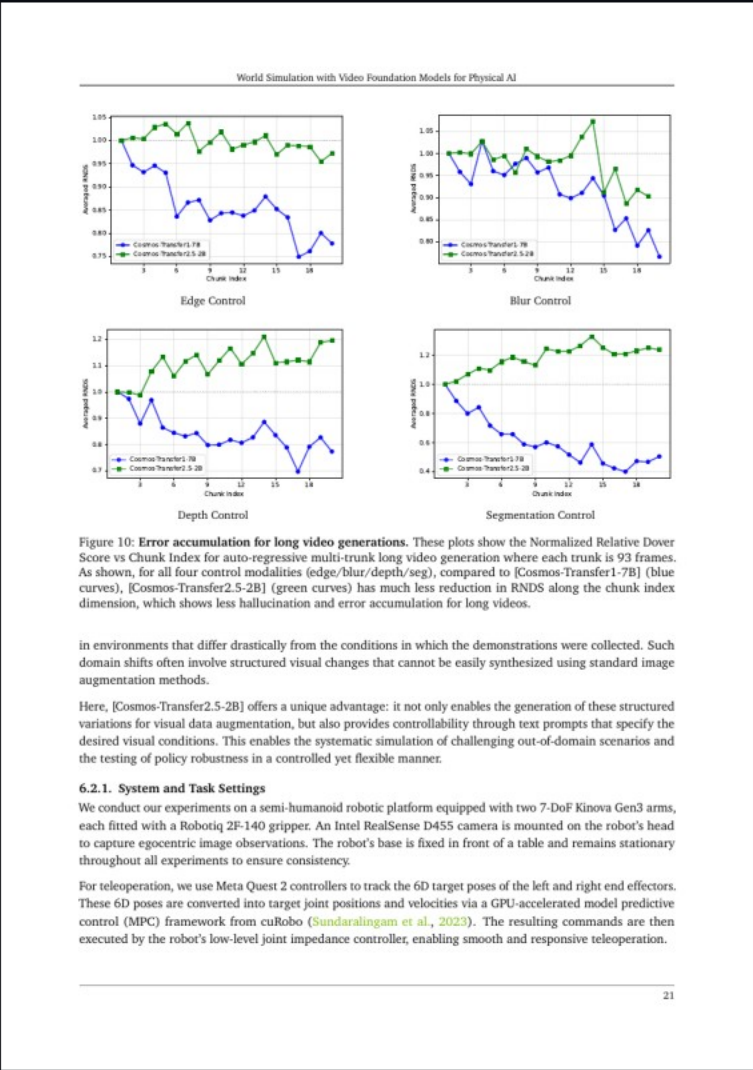
Clean ablate: TimeEmb > Cross > Concat.

Paradigm honesty (strength)

Paper §7 correctly splits latent vs pixel WM paradigms and cites V-JEPA 2 as the other pole. Reviewer should reward this and still insist applications text not blur the split.

Fig. 10 — long-horizon RNDS (Transfer)

Curves, not a single failure-rate table — cite carefully



17 videos 30–120 s; Transfer2.5 less RNDS drop vs Transfer1.

Missing experiments a reviewer wants

Not in paper — mark as gaps

G1 Head-to-head vs latent WMs (Dreamer/JEPA/LeWM) at any shared control task — §7 prose only.

G2 Matched FM vs EDM retrain on same data/budget for Predict1→2.5 attribution.

G3 Multi-task / BEHAVIOR-style language-goal SR with Transfer augs.

G4 DreamGen-trained VLA hardware SR (not only video IF).

G5 Physics probes (VideoPhy/IntPhys-class) quantifying dynamics fidelity.

G6 Published total GPU-hours for full 2B/14B pretrain (MFU snapshot $\neq$ budget).

VERDICT

Systems WFM — generative simulator not latent planner SOTA

ACCEPT as systems + applications WFM

Strong open Physical AI video world-foundation contribution. Method upgrades concrete; eval broad; paradigm split honest.

Scientifically place as GENERATIVE world simulator / synth+eval appliance — not as latent planner SOTA, not as JEPA successor.

CONDITIONS

- Keep Tab.13 scoped ·
- Label DreamGen IF $\neq$ VLA SR ·
- Distinguish offline aug vs online MPC ·
- Acknowledge proprietary curation limits

One-liner: Accept-as-systems-WFM; demand precise claims on what is synth-data vs what is online planning.

Numbers tattoo — pin these (KanBot must-cites)

Shared spine across all five seats · paper numbers only

24/30

Real2Real robot vs 5/30

0.768

PAI T2W Overall post

0.810

PAI I2W Overall post

24.95

Bridge AC PSNR

IF $\neq$ SR

DreamGen Tab.18 $\neq$ VLA

0.394

AV LET-AP Transfer2.5

3.5×

Transfer smaller vs T1

200M

curated video clips

2B / 14B

Predict scales

93f @16fps

~5.8 s generation

PARALLEL

not JEPA child

GENERATIVE

pixel WFM pole

Scientific Reviewer — close

Takeaways + exam bite

1. Strengths: open WFM narrative, concrete FM/Reason1/RL stack, broad eval, honest latent vs pixel split.
2. Weaknesses: systems density, small robot n, DreamGen $\neq$ VLA SR, proprietary data, Wan fairness.
3. Verdict: systems generative simulator — not latent planner SOTA.
4. Punish category errors that fold Cosmos into JEPA or upgrade offline aug to online MPC.

EXAM BITE

State the verdict in one sentence and name one claim that must stay scoped (Tab.13 or DreamGen IF).